PROJECT PROPOSAL

Facial Emotion Recognition Using Machine Learning, Deep Learning, and Computer Vision

Supervisor: Sir Aqib Mehdi

Dataset: FER-2013 (Kaggle) | Language: Python | Tools: OpenCV, scikit-learn, CNN

Group Member	Name	Focus Area
Member 1	Mugheeza	Machine Learning
Member 2	Kainat Ashraf	Deep Learning
Member 3	Wardah Amir	Computer Vision

1. Problem Statement

Facial expressions give important information about a person's emotional state. However, identifying emotions manually becomes difficult when there are many images or a real-time video stream. The aim of this project is to build a Facial Emotion Recognition system that automatically detects a face and classifies the expression into one of seven emotions: Angry, Disgust, Fear, Happy, Sad, Surprise and Neutral.

The project studies three approaches on the same problem: **Machine Learning (ML)**, **Deep Learning (DL)** and **Computer Vision (CV)**.

2. Motivation

Emotion recognition is an important area of Artificial Intelligence because emotions can improve the interaction between humans and intelligent systems. It can be useful in human-computer interaction, educational applications, customer feedback analysis, entertainment and assistive technologies. The project also gives practical experience in comparing traditional Machine Learning with modern Deep Learning, and in using Computer Vision for face detection and image processing.

3. Objectives

No.	Area	Objective
1	CV	Preprocess the FER-2013 images (normalization, resizing where required, augmentation) and detect faces in images and video using OpenCV.
2	ML	Extract HOG features and compare SVM, KNN and Random Forest, aiming for at least 45% test accuracy.
3	DL	Develop and train a Convolutional Neural Network (CNN) directly on the images, aiming for at least 60% test accuracy.
4	All	Compare all models using accuracy, precision, recall, F1-score and confusion matrices, and select the best one.
5	App	Build a simple application where the user gives a face image and receives the predicted emotion.

4. Proposed AI Solution

The same topic, Facial Emotion Recognition, is solved with three separate implementations. Computer Vision (OpenCV, Haar Cascade) finds and prepares the face. Machine Learning extracts HOG features and trains SVM, KNN and Random Forest classifiers. Deep Learning trains a CNN that learns features automatically. The results are compared, and the best model is used in the final application.

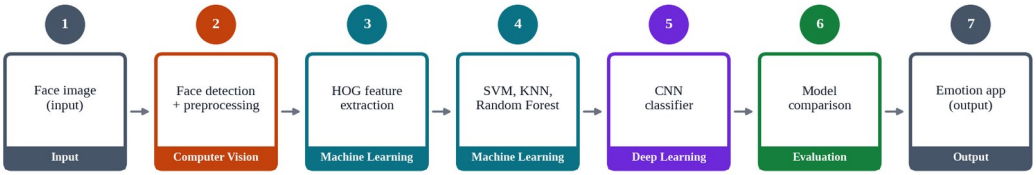


Figure 1: Overall pipeline of the proposed system

5. Dataset

Name	FER-2013 (Facial Expression Recognition 2013)
Source	kaggle.com/datasets/msambare/fer2013
Research paper	Khaireddin & Chen (2021), "Facial Emotion Recognition: State of the Art Performance on FER2013", arXiv: arxiv.org/abs/2105.03588
Images	35,887 grayscale facial images, 48 × 48 pixels
Split	28,709 training images and 7,178 testing images
Classes (7)	Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral
Preprocessing	Normalization, resizing where required, data augmentation for the CNN, and label conversion

6. Proposed Models and Techniques

Area	Model / Technique	Purpose
Computer Vision	OpenCV Haar Cascade	Face detection from images and video
Machine Learning	HOG + SVM	Emotion classification from extracted features
Machine Learning	HOG + KNN	Traditional classifier for comparison
Machine Learning	HOG + Random Forest	Additional ML comparison
Deep Learning	Convolutional Neural Network	Automatic feature learning and classification

Performance of all models is measured with accuracy, precision, recall, F1-score and confusion matrices.

7. Division of Responsibilities

Member	Area	Main responsibility
Mugheeza	Machine Learning	Data preprocessing, HOG feature extraction, SVM / KNN / Random Forest implementation and evaluation
Kainat Ashraf	Deep Learning	CNN development, training, evaluation, model saving and the emotion recognition application
Wardah Amir	Computer Vision	OpenCV face detection, image and video processing, integration and evaluation

All members contribute to the documentation, testing, presentation and final report.

8. Expected Outcome

The expected result is a working Facial Emotion Recognition system with separate ML, DL and CV parts:

- **Machine Learning:** feature-based classification using HOG with SVM, KNN and Random Forest.
- **Deep Learning:** a CNN emotion classifier and a simple prediction application.
- **Computer Vision:** face detection and processing from images or video.

The project will also show the strengths and limitations of each approach.

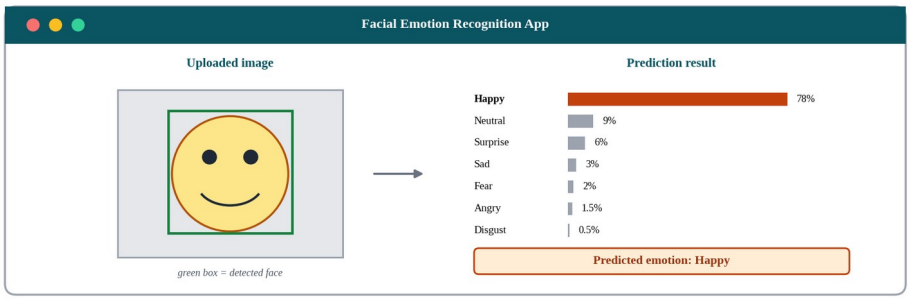


Figure 2: Concept of the application output (illustrative values only)

9. Conclusion

This project proposes a complete Facial Emotion Recognition system using Machine Learning, Deep Learning and Computer Vision on the FER-2013 dataset. The final comparison will show which approach works best for recognizing emotions from facial images.